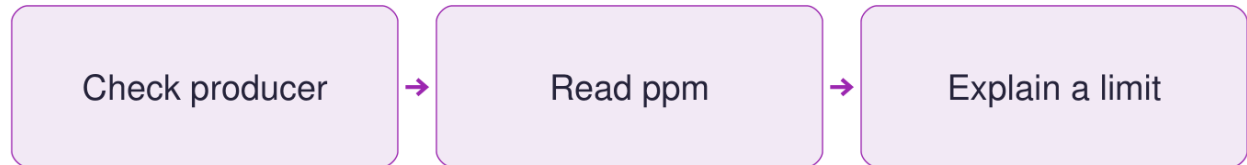


Resource use and carbon dioxide

Pupil resources | LAUNCH Humanities Summer 1 Week 1

Explain resource use and climate change using online data.



Authored teaching picture

Use the evidence cards and blank spaces to give your own response. Speak, point, draw, type or write.

Start here: provenance means who made a source, when and why. Read cards A and D for units and values; no earlier lesson is needed.

Evidence cards

Check the source and limit before using a detail

A The measured record

NOAA Global Monitoring Laboratory publishes the Mauna Loa annual mean CO₂ series to monitor atmospheric change. File created 5 September 2026; accessed 21 September 2026. The selected values run from 338.76 ppm in 1980 to 427.35 ppm in 2025.

Source: NOAA GML annual mean CO₂ file; 2026 release

Limit: One observatory record is not the global average. The selected years omit changes between them; values can be revised.

B Explaining the cause

NASA explains that burning fossil fuels releases CO₂ and that greenhouse gases slow heat loss to space. This connects resource use for energy with climate change. NASA made this public science explanation; page updated 13 July 2026, read 21 September 2026.

Source: NASA Causes; public explanation, updated 2026

Limit: This explanation supports a mechanism. The selected CO₂ table is not a temperature record.

C A weaker claim

AUTHORED EXAMPLE created for this lesson on 21 September 2026: an unattributed post says CO₂ figures prove every place warmed by the same amount. Its imagined purpose is to persuade, but it supplies no author, measurement method or evidence.

Source: Authored source-critique example; not a real post

Limit: A confident claim is not a measurement. Do not use this invented post as evidence about climate.

D What the unit measures

NOAA states that CO₂ is expressed as a mole fraction in dry air, abbreviated ppm. Read ppm as parts per million. The producer is NOAA GML; the file dated 5 September 2026 makes monitoring data available for scientific and public use.

Source: NOAA GML file header; accessed 21 September 2026

Limit: Concentration is evidence about a cause of warming, not a reading of temperature. Keep those quantities separate.

NOAA Mauna Loa annual mean CO2 selected years

Selected annual means

Units: ppm (parts per million), dry air

Year	CO2 ppm
1980	338.76
1990	354.45
2000	369.71
2010	390.10
2020	414.21
2025	427.35

Source: NOAA GML annual mean CO2 file, created 5 September 2026; accessed 21 September 2026.

https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_annmean_mlo.csv

One observatory is not the global average. CO2 concentration is evidence of a cause, not a measure of temperature. Selected years omit intervening values; data can be revised.

My annotations: producer, date and purpose

A pattern using two values

Arrival task

Supported

Start here: provenance means who made a source, when and why. Read cards A and D for units and values; no earlier lesson is needed.

1 Who produced card A?

Help: Read its source line.

My response

2 Which unit is printed for CO₂?

Help: Use the table heading and card D.

My response

3 Which selected year has the highest value?

Help: Compare the six rows.

My response

4 Can this table give the air temperature?

Help: Read card D.

My response

Arrival task

Standard

Start here: provenance means who made a source, when and why. Read cards A and D for units and values; no earlier lesson is needed.

1 What does provenance ask us to check?

Help: Use the glossary.

My response

2 Name the first and last selected values.

Help: Read the endpoint rows.

My response

3 How does NASA link fossil fuels to CO₂?

Help: Read card B.

My response

4 Why is card C weaker for a factual claim?

Help: Check its authorship and evidence.

My response

Arrival task

Stretch

Start here: provenance means who made a source, when and why. Read cards A and D for units and values; no earlier lesson is needed.

1 Why does a clear source date matter?

Help: Compare an observation year with file date.

My response

2 Does a rise between selected rows describe every intervening year?

Help: Look at the gaps in years.

My response

3 Can one observatory equal a global average?

Help: Read the limit on card A.

My response

4 What extra evidence would test a temperature claim?

Help: Name the quantity needed.

My response

Shared investigation

Use the evidence; explain and check together

1. Read cards A to D. Check producer, date and purpose, then annotate the six NOAA rows and distinguish the authored weak source.
2. Plot the checked table with year and ppm labels. Describe its pattern, connect fossil-fuel use with CO₂ using NASA, and state a limit.

Horizontal axis / vertical axis / units

Pattern using two checked values

Resource use, CO₂ and the source of my explanation

A limitation of the evidence

Independent task

Supported

Explain resource use and climate change using online data.

1. Read the highlighted 1980 and 2025 rows and label the units with the word bank.
2. Check the plotted values with an adult reader, then annotate producer and date.
3. Independently explain one resource-use link using card B and one limit.

Help: Use highlighted evidence and the frame. An adult may read or scribe your own words.

Producer date purpose and unit

Pattern with two checked values

Resource-use explanation and source

Evidence limit and next check

Further challenge: Explain a second detail if ready.

Independent task

Standard

Explain resource use and climate change using online data.

1. Annotate producer, date, purpose and units for the NOAA table.
2. Plot all six rows and describe a pattern with two values.
3. Independently explain the resource-use link and compare the usefulness of cards A and C.

Help: Use the cards and organiser. Choose your own evidence and explanation.

Producer date purpose and unit

Pattern with two checked values

Resource-use explanation and source

Evidence limit and next check

Further challenge: Compare your reasoning with one alternative.

Independent task

Stretch

Explain resource use and climate change using online data.

1. Audit the selected years, provenance and the distinction between concentration and temperature.
2. Plot all six rows; explain what the missing intervening years could conceal.
3. Use NASA to explain the mechanism without claiming the table alone proves every impact.
4. Independently justify the stronger source and propose a further evidence check.

Help: Use the source limits and test a different priority or assumption.

Producer date purpose and unit

Pattern with two checked values

Resource-use explanation and source

Evidence limit and next check

Further challenge: Propose a further evidence check and explain its value.

Exit tickets

Use your chosen route and give your own response

Supported

1. Name a selected CO₂ value with its unit. Use a named detail and your own explanation.

2. Give one limit of that record. Use a named detail and your own explanation.

Standard

1. Explain the resource-use link with a source. Use a named detail and your own explanation.

2. What can the selected table not establish? Use a named detail and your own explanation.

Stretch

1. Why is your chosen evidence more useful than the weak example? Use a named detail and your own explanation.

2. Which further check would strengthen your explanation? Use a named detail and your own explanation.
